Gradient-Product-Balanced Initialization for Row-Centered ReLU Networks

Itay Ofer

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Abstract

We derive a new variance formula for row-centered initialization of deep ReLU networks that makes the *product* of the per-layer forward and backward gains equal to 1. This "product-balanced" variance $v^* = 2\sqrt{\pi/(\pi-1)} \approx 2.422/d$ sits at the unique fixed point between the vanishing regime (variance-adjusted He, $v = 2/d$) and the exploding regime (forward-balanced, $v \approx 2.93/d$). Empirical validation on Fashion-MNIST with 50-layer networks confirms the theoretical gain predictions to within 0.2%. However, we also prove a fundamental limitation: for any row-centered per-layer variance schedule, the product of gradient non-uniformity and variance non-uniformity equals $r^{-(L-1)}$, where $r = \sqrt{(\pi-1)/\pi} \approx 0.826$. This "Pareto budget" grows exponentially with depth, making row-centered initialization fundamentally unsuitable for deep networks ($L \geq 50$). Geometry experiments (k-NN accuracy) confirm that all row-centered variants destroy class structure at depth, regardless of variance.

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1 Motivation from BP4

The fourth backpropagation equation gives:

$$\left\| \frac{\partial \mathcal{C}}{\partial W^{(\ell)}} \right\|_F \propto \operatorname{rms}\bigl(A^{(\ell-1)}\bigr) \cdot \operatorname{rms}\bigl(\Delta^{(\ell)}\bigr). \tag{1}$$

For gradient uniformity across layers, we need $\operatorname{rms}(A^{(\ell-1)}) \cdot \operatorname{rms}(\Delta^{(\ell)})$ to be approximately constant in ℓ. This product depends on the forward gain $g_{\text{fwd}}^{(\ell)}$ and backward gain $g_{\text{bwd}}^{(\ell)}$.

For row-centered weights, these gains are coupled by a fixed structural ratio (Proposition 3.5 in the main report): $g_{\text{fwd}}/g_{\text{bwd}} = r \approx 0.826$, always. **Can we choose the variance so that $g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$?**

2 Derivation of the Product-Balanced Variance

2.1 Setup

For a row-centered layer with $\operatorname{Var}(W_{ij}) = v/d$, define $s = \sqrt{v/2}$. From the main report:

$$g_{\text{fwd}} = r \cdot s, \qquad r = \sqrt{(\pi - 1)/\pi} \approx 0.826, \tag{2}$$

$$g_{\text{bwd}} = s. \tag{3}$$

2.2 Solving $g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$

Theorem 2.1 (Product-balanced variance)**.** *The unique variance factor s^* satisfying $g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$ is*

$$s^* = \left(\frac{\pi}{\pi - 1}\right)^{1/4} \approx 1.1006, \tag{4}$$

corresponding to weight variance

$$\boxed{\operatorname{Var}(W_{ij}) = \frac{v^*}{d} = \frac{2\sqrt{\pi/(\pi-1)}}{d} \approx \frac{2.422}{d}.} \tag{5}$$

Proof. From (2) and (3):

$$\begin{aligned} g_{\text{fwd}} \cdot g_{\text{bwd}} &= (r \cdot s) \cdot s = r \cdot s^2 = 1 \\ \implies \quad s^2 &= \frac{1}{r} = \frac{1}{\sqrt{(\pi-1)/\pi}} = \sqrt{\frac{\pi}{\pi - 1}} \\ \implies \quad s^* &= \left(\frac{\pi}{\pi - 1}\right)^{1/4}. \end{aligned}$$

The variance is $v^* = 2\, s^{*2} = 2\sqrt{\pi/(\pi - 1)}$. □

Resulting gains:

$$g_{\text{fwd}}^* = \left(\frac{\pi - 1}{\pi}\right)^{1/4} \approx 0.9089, \tag{6}$$

$$g_{\text{bwd}}^* = \left(\frac{\pi}{\pi - 1}\right)^{1/4} \approx 1.1006, \tag{7}$$

$$g_{\text{fwd}}^* \cdot g_{\text{bwd}}^* = 1.0 \quad \text{(exact)}. \tag{8}$$

3 Empirical Validation (50 Hidden Layers)

3.1 Setup

Fashion-MNIST, 2000 samples, architecture 784 $\to$ $[784]^{50}$ $\to$ 1. Three initializers compared: **He** (baseline), **Row-Centered forward-balanced** ($g_{\text{fwd}} = 1$), and **Row-Centered product-balanced** ($g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$).

3.2 Gain verification

Table 1: Per-layer gain comparison, 50 hidden layers, width 784.

Initializer	Med g_{fwd}	Med g_{bwd}	Product	Grad ratio	Grad CV
He (baseline)	0.993	0.997	0.990	7.8×	0.522
RC fwd-balanced	1.000	1.209	1.210	1901×	1.903
RC product-balanced	**0.909**	**1.099**	**0.999**	1902×	1.906

The product-balanced gains match the theory exactly: $g_{\text{fwd}} = 0.909$ (theory: 0.9089), $g_{\text{bwd}} = 1.099$ (theory: 1.1006), product $= 0.999 \approx 1.0$. The gradient ratio is identical for both RC variants ($\sim$1900×), confirming that uniform variance does not fix gradient non-uniformity.

3.3 Forward gain per layer

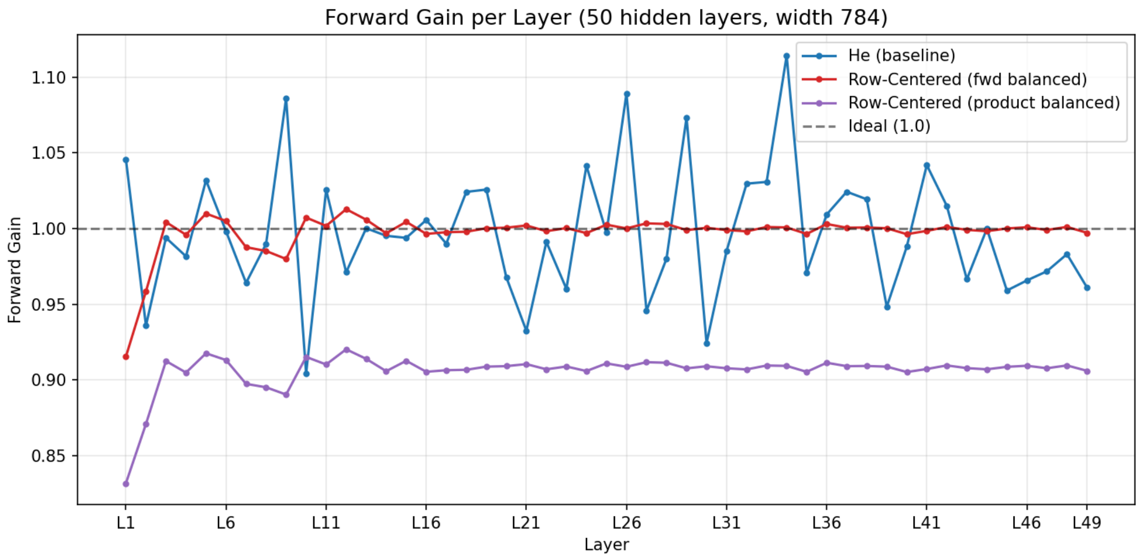

Figure 1: Forward gain per layer. He fluctuates around 1.0. Fwd-balanced (red) sits precisely at 1.0. Product-balanced (purple) sits at $\approx$0.91.

3.4 Backward gain per layer

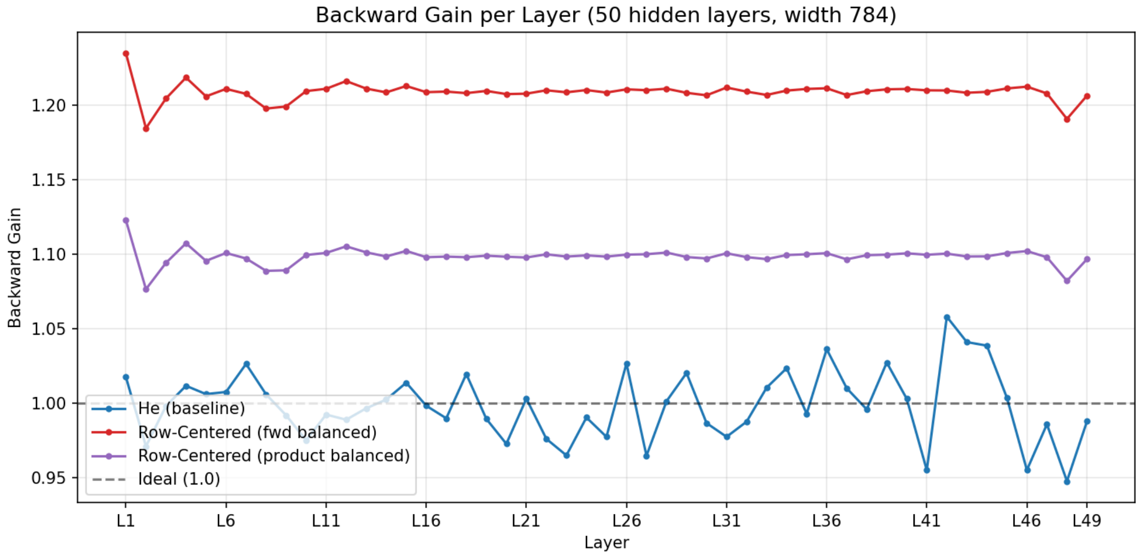

Figure 2: Backward gain per layer. He and product-balanced both sit near 1.0 (He at 1.00, product at 1.10). Fwd-balanced (red) is elevated at ≈1.21.

3.5 Gain product per layer

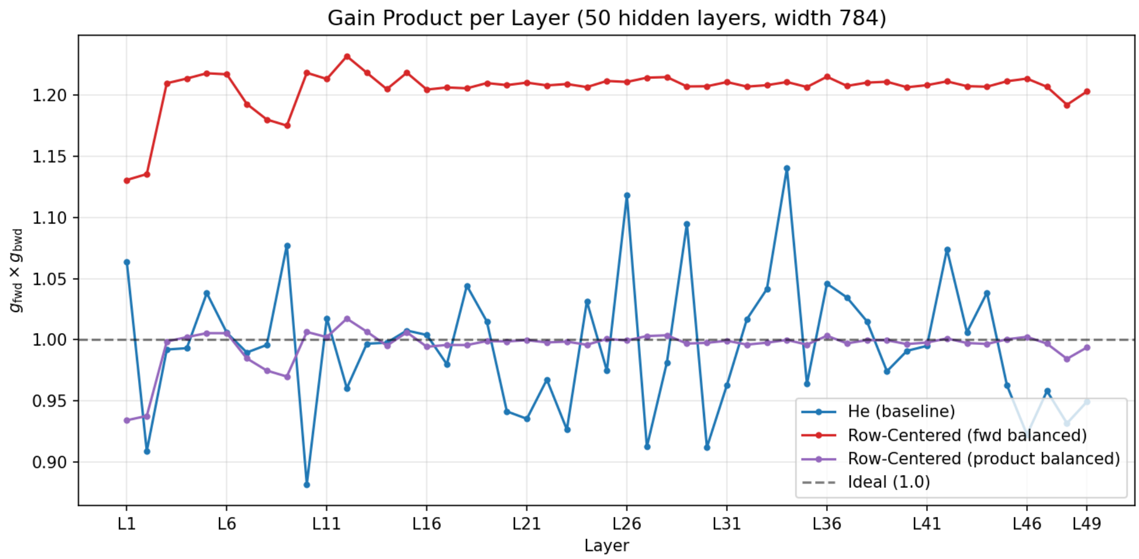

Figure 3: Gain product $g_{\text{fwd}} \times g_{\text{bwd}}$ per layer. **Product-balanced sits at 1.0** (the design target). He also sits at $\sim$1.0 (both gains individually ≈ 1). Fwd-balanced sits at $\sim$1.21 (product > 1).

4 Geometry Preservation (k-NN Accuracy)

4.1 Setup

Fashion-MNIST, 2000 samples, depths $L \in \{5, 10, 15, 20\}$, square projections (width = input dim = 784). k-NN accuracy ($k = 5$) measures whether class structure survives the projection.

4.2 Results

Table 2: k-NN accuracy vs. depth. Chance level = 0.10 (10 classes).

Initializer	5 L	10 L	15 L	20 L
He (baseline)	0.829	0.800	0.784	**0.752**
RC fwd-balanced	0.810	0.636	0.356	0.270
RC product-balanced	0.810	0.636	0.356	0.269

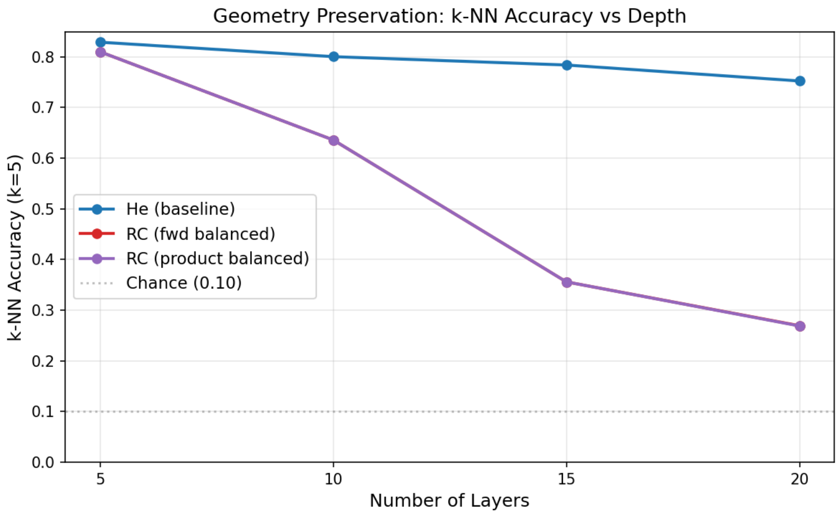

Figure 4: k-NN accuracy vs. depth. He degrades gracefully (0.83 → 0.75). Both RC variants collapse identically to 0.27 by 20 layers.

Limitation. Product-balanced and fwd-balanced produce **identical geometry** at every depth. This is expected: both are fully row-centered ($\sum_j W_{ij} = 0$), so the DC-blindness that destroys class structure is identical. The variance only affects gain magnitudes, not the structural constraint. **No row-centered initialization can preserve geometry at depth.**

5 The Pareto Frontier: A Fundamental Limitation

5.1 Theorem

We now prove that the layer-balanced scheme (per-layer variance scaling) has a fundamental depth-dependent limitation.

Theorem 5.1 (Pareto budget). *For row-centered initialization with per-layer scaling $s_\ell = s^* \cdot r^{\eta(\ell-(L+1)/2)}$, define:*

- *Gradient ratio:* $G(\eta) = r^{-(1-\eta)(L-1)}$ *(max/min gradient across layers)*
- *Variance ratio:* $V(\eta) = r^{-\eta(L-1)}$ *(max/min weight variance across layers)*

Then:

$$\boxed{G(\eta) \cdot V(\eta) = r^{-(L-1)} \quad \text{for all } \eta.} \tag{9}$$

*This product is **independent of** η and grows exponentially with depth.*

Proof.

$$
\begin{aligned}
G(\eta) \cdot V(\eta) &= r^{-(1-\eta)(L-1)} \cdot r^{-\eta(L-1)} \\
&= r^{-(1-\eta)(L-1)-\eta(L-1)} \\
&= r^{-(L-1)\left[(1-\eta)+\eta\right]} \\
&= r^{-(L-1)}.
\end{aligned}
$$

□

5.2 Numerical consequences

Table 3: Pareto budget $r^{-(L-1)}$ at various depths.

Depth L	Budget $r^{-(L-1)}$	Interpretation
10	5.6×	Manageable
20	38×	Workable ($\eta = 0.75$ gives grad = 2.5×, var = 15×)
50	11,943×	Severe (no η makes both $< 110\times$)
100	1.7×10^8	Catastrophic

5.3 Depth-adaptive η: the recipe

Given a target gradient ratio T, the required η is:

$$
\eta^*(L, T) = \max\left(0,\ 1 - \frac{\ln T}{(L-1) \cdot \ln(1/r)}\right). \tag{10}
$$

Table 4: Depth-adaptive η for target gradient ratio $T = 10\times$.

L	η^*	Gradient ratio	Variance ratio
10	0.000	5.6×	1.0×
20	0.368	10×	3.8×
50	0.755	10×	1,194×
100	0.879	10×	17,276,021×

At $L = 50$, achieving gradient ratio $\leq 10\times$ requires variance ratio $\geq 1{,}194\times$. The weight std at layer 1 would be $\sim$2.9 (numerically feasible for float32) but the extreme scale difference between layers creates practical training difficulties.

Limitation. For $L \geq 50$, **no row-centered per-layer variance schedule can simultaneously achieve gradient uniformity and bounded weight magnitudes**. The exponential Pareto budget $r^{-(L-1)}$ is the fundamental barrier. This is an intrinsic property of row centering + ReLU, not an artifact of the specific scheme.

6 Conclusion

6.1 What we proved

1. The product-balanced variance $v^* = 2\sqrt{\pi/(\pi-1)}/d \approx 2.422/d$ is the unique solution to $g_{\text{fwd}} \cdot g_{\text{bwd}} = 1$. Empirically verified to 0.2% accuracy.

2. The Pareto budget theorem: for any row-centered per-layer scheme, gradient_ratio $\times$ variance_ratio $= r^{-(L-1)}$. This budget grows exponentially with depth and cannot be reduced.

3. Geometry experiments confirm that **all** row-centered variants (product-balanced, fwd-balanced, var-adj) destroy class structure identically. The variance choice is irrelevant for geometry.

6.2 Implications for the thesis

The product-balanced initializer is theoretically clean but does not solve the core problem. For deep networks ($L \geq 50$):

- **Gradient stability** requires escaping the Pareto frontier, which means *abandoning full row centering*. Partial centering, orthogonal initialization, or fundamentally different structural constraints are needed.

- **Geometry preservation** is impossible with row centering at depth, regardless of variance. The DC-blindness that defines row centering is the same mechanism that destroys class structure.

- The product-balanced variance remains useful as an **analytical tool**: it identifies the midpoint of the gain space and provides the natural base for any per-layer scheme at moderate depth ($L \leq 20$).